

Malin Ouyang

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Education:

- Nanjing University, Nanjing, China
Bachelor of Science in Astrophysics
September 2022 – July 2026
- No.3 Middle School of Kunming, Kunming, China
High school student
September 2019 – September 2022

Research Experience:

National Natural Science Foundation Youth Student Basic Research Program, Nanjing University – School of Astronomy and Space Science

Nanohertz Gravitational Waves—A New Probe of the Matter Distribution of the Milky Way

October 2023 – Present

- Investigated how the Milky Way matter distribution influences the propagation of nanohertz gravitational waves.
- Applied numerical methods to improve the fit to the Hellings-Downs curve.
- Contributed to developing models to constrain the Milky Way matter distribution.

Undergraduate Innovation Program, ICTP-AP Gravitational Wave Cosmology Tai Chi Laboratory (Beijing)

Detection of Gravitational Waves using Deep Learning

June 2024 – January 2025

- Developed machine learning models (CNN, ResNet) for detecting gravitational wave signals.
- Focused on binary neutron star (BNS) and binary black hole (BBH) systems.
- Collaborated with the Gravitational Wave Data Analysis and Machine Learning Research Group.

Visiting Research Student, Ludwig-Maximilians-Universität München (Germany)
Hunting Optical Counterparts of Gravitational Wave Events with Wendelstein
Observatory

February 2025 – July 2025

- Conducted real-time follow-up observations of LIGO/Virgo/KAGRA O4 gravitational wave events.
- Built modular pipelines for photometric data analysis, light curve modeling, and signal-to-noise ratio computation.
- Assessed detectability of BNS, NSBH, BBH, and AGN events.

Publications and Patents:

(No publications or patents yet)

Awards, Interests, Skills, and Community Involvement:

Skills:

- Programming: C, C++, Python
- Tools: LaTeX, Linux, Jupyter Notebook
- Languages: Chinese (native), English (fluent)

Community Involvement & Leadership:

- Deputy Secretary, Youth League Committee, School of Astronomy and Space Science (2024–Present)
- Secretary, Youth League Branch, Class of 2022 (2023–Present)
- Volunteer teaching in rural Yunnan; awarded Outstanding Social Practice Team (Jiangsu Province) and Top Ten in National “Research China” Competition

Awards:

- National Natural Science Foundation Youth Student Basic Research Program (Undergraduate), Only 15 students university-wide each year
- Nanjing University Dongliang Excellence Scholarship (First Class), The highest honor for undergraduate students of Nanjing University, with less 13 awardees each year
- National Astronomical Observatories, CAS Scholarship
- Nanjing University Undergraduate Fundamental Discipline Special Scholarship
- National Astronomical Observatories, CAS Scholarship
- Nanjing University Dongliang Excellence Scholarship
- Outstanding Individual in Social Practice, Nanjing University
- Outstanding Communist Youth League Member, Nanjing University
- Outstanding Minority Student, Nanjing University (2023-2024 Academic Year)
- Outstanding Student, Nanjing University (2024 Academic Year)